

WEEK 2 REPORT

Week 2 Report - Cybersecurity Internship

User Management System Security Implementation

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Role: Cybersecurity Intern

Project: Secure User Authentication System

1. OBJECTIVE

The objective of this week was to implement security features in a User Management System, including authentication, password protection, and secure API access using JWT tokens.

2. PROJECT OVERVIEW

This project is a backend-based User Management System developed using Node.js, Express, and MongoDB.

In Week 2, the focus was on improving the security of the system by implementing authentication and protecting sensitive routes.

3. TASKS COMPLETED

3.1 Password Security

- Implemented bcrypt library for password hashing
- Passwords are now stored in hashed format in MongoDB
- Plain text password storage removed

3.2 Authentication System

- Implemented JWT (JSON Web Token) authentication
- Token generated on successful login
- Token used for verifying user identity

3.3 Protected Routes

- Created authentication middleware (verifyJWT)
- Protected sensitive routes like /account
- Only valid token users can access protected data

3.4 API Testing (Postman)

- Tested user registration API
- Tested login API and token generation
- Tested protected route with Authorization header
- Tested invalid token response (401 Unauthorized)

4. SECURITY IMPROVEMENTS

- Password hashing using bcrypt
- JWT-based authentication system
- Middleware protection for secure routes
- Input validation added in backend

5. RESULTS

- User registration working successfully
- Login returns valid JWT token
- Protected routes accessible only with valid token
- MongoDB stores hashed passwords securely

6. TOOLS USED

- Node.js
- Express.js
- MongoDB
- JWT (jsonwebtoken)
- bcrypt
- Postman

7. SCREENSHOTS INCLUDED

- User registration success response
- Login response with token
- Protected route response (/account)
- MongoDB user data (hashed password)

8. CONCLUSION

Week 2 focused on implementing secure authentication in the system.

The application is now protected using JWT tokens and passwords are securely hashed using bcrypt.

The system is now more secure and ready for advanced testing in Week 3.